

PROBLEMS SUBMIT CODE MY SUBMISSIONS STATUS STANDINGS CUSTOM INVOCATION

D. Count Paths

time limit per test: 1.5 seconds
memory limit per test: 256 megabytes

You are given a graph with n vertices (numbered 1 through n) and m **directed** edges. Count paths consisting of k steps (edges) and print the answer modulo $10^9 + 7$. A path can visit the same vertex or edge multiple times.

Input

The first line contains three integers n , m and k ($1 \leq n \leq 100$, $0 \leq m \leq n \cdot (n - 1)$, $1 \leq k \leq 10^9$) — the number of vertices, the number of edges and the length of a path.

The following m lines describe edges. The i -th of these lines contains two integers a_i and b_i ($1 \leq a_i, b_i \leq n$, $a_i \neq b_i$), denoting a directed edge from a_i to b_i . There are no self-loops or multiple edges.

Output

Print the answer modulo 1000000007.

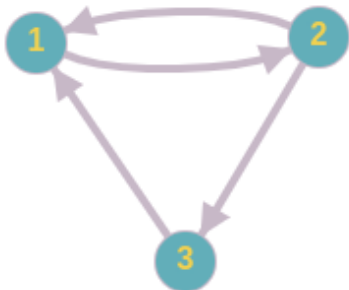
Examples

input	Copy
3 4 2 1 2 2 3 3 1 2 1	
output	Copy
5	

input	Copy
5 10 11 2 3 4 2 2 1 2 4 1 5 5 2 3 2 3 1 3 4 1 2	
output	Copy
21305	

Note

In the first sample test, we are given the following graph with $n = 3$ vertices and $m = 4$ edges.



There are 5 paths of length $k = 2$:

- $1 \rightarrow 2 \rightarrow 1$
- $1 \rightarrow 2 \rightarrow 3$
- $2 \rightarrow 1 \rightarrow 2$
- $2 \rightarrow 3 \rightarrow 1$
- $3 \rightarrow 1 \rightarrow 2$

Matrix Exponentiation

Finished

Practice



→ Virtual participation

Virtual contest is a way to take part in past contest, as close as possible to participation on time. It is supported only ICPC mode for virtual contests. If you've seen these problems, a virtual contest is not for you - solve these problems in the archive. If you just want to solve some problem from a contest, a virtual contest is not for you - solve this problem in the archive. Never use someone else's code, read the tutorials or communicate with other person during a virtual contest.

Start virtual contest

→ Clone Contest to Mashup

You can clone this contest to a mashup.

Clone Contest

→ Submit?

Language: GNU G++23 14.2 (64 bit, ms)
Choose file: Choose File No file chosen
Submit

→ Last submissions

Submission	Time	Verdict
336302028	Aug/30/2025 21:38	Happy New Year!
336301880	Aug/30/2025 21:36	Happy New Year!
336287470	Aug/30/2025 19:45	Happy New Year!
336240750	Aug/30/2025 13:14	Happy New Year!
336180827	Aug/29/2025 22:04	Happy New Year!
336179545	Aug/29/2025 21:50	Happy New Year!
336179383	Aug/29/2025 21:49	Happy New Year!